OL	T	H	W	R
1) Sunny	Hot	High	Weak	No
2) Sunny	Mild	High	Strong	No
3) Cloudy	Hot	High	Weak	Yes
4) Rainy	Mild	High	Weak	Yes
5) Rainy	Cool	Normal	Weak	Yes
6) Rainy	Cool	Normal	Strong	No
7) Cloudy	Mild	Normal	Strong	Yes
8) Sunny	Cool	Normal	Weak	No

Root entropy:

$$\text{Entropy} = i \sum pi \log_2 (pi)$$

Dataset:

	Total	Yes	No
R	8	4	4

Calculation:

$$= -\frac{4}{8} \log_2 \frac{4}{8} - \frac{4}{8} \log_2 \frac{4}{8}$$

$$= -\left[0.5 \log_2 0.5 + 0.5 \log_2 0.5\right]$$

$$= 1$$

Temperature:

	Total	Yes	No	Entropy
Hot	2	1	1	1
Mild	3	2	1	0.918
Cool	3	1	2	0.918

Calculation:

$$\text{Hot} = -\left[\frac{1}{2}\log_2\frac{1}{2} + \frac{1}{2}\log_2\frac{1}{2}\right] = 1$$

$$\text{Mild} = -\left[\frac{2}{3}\log_2\frac{2}{3} + \frac{1}{3}\log_2\frac{1}{3}\right] = 0.918$$

$$\text{Cool} = -\left[\frac{1}{3}\log_2\frac{1}{3} + \frac{2}{3}\log_2\frac{2}{3}\right] = 0.918$$

Weighted Entropy of T:

$$= \frac{2}{8}(1) + \frac{3}{8}(0.918) + \frac{3}{8}(0.918)$$

$$= 0.9385$$

Information Gain:

$$IG = 0.0615$$

Humidity dataset:

	Total	Yes	No	Entropy
High	4	2	2	1
Normal	4	2	2	1

Calculation:

$$\text{High} = -\left[\frac{2}{4}\log_2\frac{2}{4} + \frac{2}{4}\log_2\frac{2}{4}\right] = 1$$

$$\text{Low} = -\left[\frac{2}{4}\log_2\frac{2}{4} + \frac{2}{4}\log_2\frac{2}{4}\right] = 1$$

Weighted Entro

$$= \frac{4}{8}(1)$$

~~IG = 1 - 1~~

Information G

= 1

= 0

Wind dataset

	Tota
Weak	5
Strong	3

Calculation:

Weak = -

Strong = -[1/3

Weighted En

=

-

Information

(3)

Weighted Entropy of Humidity:

$$= \frac{4}{8}(1) + \frac{4}{8}(1) = 1$$

~~IG = 1 - 1 = 0~~

Information Gain:

$$= 1 - 1$$
$$= 0$$

Wind dataset:

	Total	Yes	No	Entropy
Weak	5	3	2	
Strong	3	1	2	

Calculation:

$$\text{Weak} = -\left[\frac{3}{5}\log_2\frac{3}{5} + \frac{2}{5}\log_2\frac{2}{5}\right] = 0.971$$

$$\text{Strong} = -\left[\frac{1}{3}\log_2\frac{1}{3} + \frac{2}{3}\log_2\frac{2}{3}\right] = 0.918$$

Weighted Entropy:

$$= \frac{5}{8}(0.971) + \frac{3}{8}(0.918)$$

$$= 0.9511$$

Information Gain:

$$= 1 - 0.9511$$
$$= 0.0489$$

Outlook Dataset

~~Outlook~~	Total	Yes	Not	Entropy
~~Sam~~ Sunny	3	0	3	
Cloudy	2	2	0	
Rainy	~~3~~	2	1	

Entropy:

Sunny: $-\left[\frac{0}{3}\log_2(0/3) + \frac{3}{3}\log_2\frac{3}{3}\right] = 1$

Cloudy: $-\left[\frac{2}{2}\log_2\frac{2}{2} + \frac{0}{2}\log_2\frac{0}{2}\right] = 1$

Rainy: $-\left[\frac{2}{3}\log\frac{2}{3} + \frac{1}{3}\log\frac{1}{3}\right] = 0.918$

Weighted Entropy:

$$= \frac{3}{8}(1) + \frac{2}{8}(1) + \frac{3}{8}(0.918)$$

$$= 1.0942$$

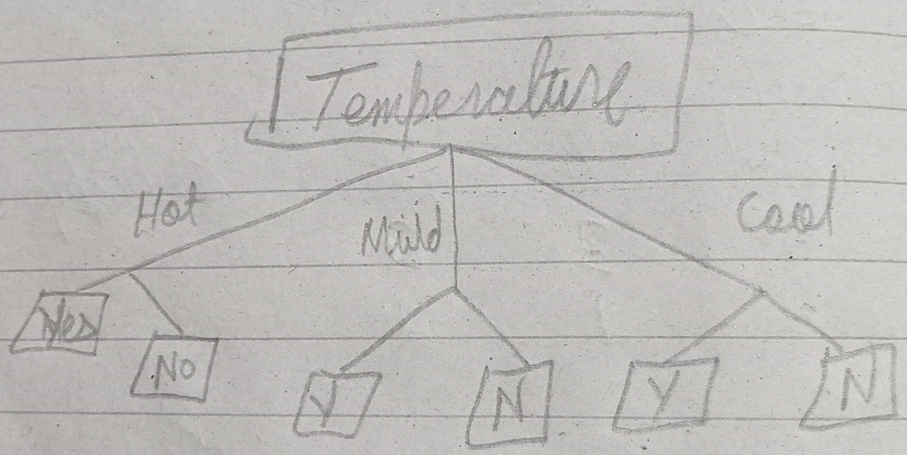
$$IG = 1 - 1.0942$$

$$= -0.094$$

(5)

	Entropy	IG
Outlook	1.094	-0.0942
Temperature	0.9385	0.0615
Humidity	1	0
Wind	0.9511	0.0489

Temperature is root node because IG is h



Outlook Dataset

~~Total~~ T

~~Some~~ Sunny

Cloudy

Rainy

Entropy:

Sunny: $-\left[\frac{0}{3}\right.$

Cloudy: $-\left[\frac{2}{2}\right.$ log

Rainy: $-\left[\frac{2}{3}\right.$

Weighted Entro

$= \frac{3}{8}$

$= 1.09$

IG = 1 -

$= -0.09$